

Provide feedback on each artifact: GitHub repo, code, code output, project report

GitHub repo:

Good Work:

Within the Spotify project Repository, all key elements were included (Data, Model, Notebook, Report). The placement of each file was intuitive, and separating the raw data from the Python and PDF files was logical.

Potential Issue:

While there were readme and requirements placeholders, the files were empty. Populating these files with project overview, instructions for running the code, and explanations would help with accessibility. However, it seems that lots of this valuable information was covered in your final report. Additionally, the notebooks in your repo could be condensed to remove the unused folders, and keeping the reports in the same format (DOX or PDF, not both) would be ideal.

Code:

Good Work:

The final notebook is easy to follow and clearly labeled. It clearly follows your engineering progression. You worked with a large dataset (8,686 rows and 26 variables) built from combining multiple playlists into a new dataframe. The data cleaning process made sense: removing duplicates, dropping unnecessary columns, and keeping only the first listed artist. Later cleaning steps, like addressing long runtime issues and filtering out non-English lyrics, improved overall performance. The 70:15:15 data split was also a solid choice for training, validation, and testing. It was great to see how you integrated lyrical, audio, and genre features into your KNN approach. The layout made the final model easy to follow.

Potential Issue:

Adding more inline comments, particularly highlighting the meaning of your code outputs, could add a lot of context to your project. It's also worth noting that not all songs had lyrics available from the API, which reduced the dataset size, and that the deduplication method of keeping only the more popular artist version of a song could introduce bias. Some of the more complex code blocks, such as the feature weighting logic and cosine similarity calculations, could also benefit from quick, plain-language explanations right above them.

Code output:**Good Work:**

The code output clearly shows how the project evolved, and you added more features to improve the model. It is clear how you evolved and adapted when faced with obstacles such as different genre tags and sparse data. The final model was complex but clearly well tested, and the features were weighted accordingly. The visualizations were a strong point. The distribution graphs were clear and easy to interpret, and the correlation matrix was well-presented, highlighting both strong positive and strong negative relationships effectively.

Potential Issue:

One challenge is that the code provided for each week is only partial, which makes it harder to follow the full progression of the project and see how each step builds on the last. While KNN is a solid starting point for this type of task, it can be slow with large datasets. Even after the dataset was cut in half before evaluation, the runtime remained long. Since KNN is a distance-based model, it may not be the most efficient or optimal choice as the final model for this type of recommendation system.

Project report:**Good Work:**

The report is very strong, follows all the rubrics, and articulates the process, backing it up with strong visuals. The inclusion of the example recommendations was helpful as well, and your table clearly outlined the process, displaying how the model improved as you feature engineered and tuned it. The data cleaning steps were rigorous, duplicates were removed, non-English songs were filtered out, and potential ethical and cultural biases were addressed. Model 10, the final version, achieved 60% test accuracy, showing clear progress and effective integration of multiple feature types.

Potential Issue:

The report is clearly well put together and thoughtfully written, so these suggestions are minimal. Adding a bit more focus on the potential end-user experience, such as how the recommendation system could be implemented, would be interesting. It could also help to elaborate on limitations, like the dataset reduction caused by missing lyrics, the potential bias from keeping only the more popular artist version of duplicate tracks, and the runtime of KNN.

Overall, great work, it was awesome to learn more about your project and the thinking behind your decision-making!